DSA0603-Data Handling and Visualizaition

Visualization Activities with datasets

NAME:MOHAN RAJ VD

REG NO: 192424151

Activity 1: Heatmap – Student Performance Analysis

Program:

```
library(pheatmap)
student <- c("S1","S2","S3","S4","S5","S6","S7","S8","S9","S10")
marks <- data.frame(
  Mathematics = c(92,76,81,65,90,58,84,71,95,79),
  Physics = c(88,72,78,70,94,62,86,75,91,81),
  Chemistry = c(84,69,83,68,91,60,82,73,93,77),
  Programming = c(95,80,85,72,96,65,88,77,98,83),
  English = c(81,74,79,75,89,70,80,72,94,78)
)
rownames(marks) <- student
pheatmap(
  marks,
  cluster_rows = FALSE,
  cluster_cols = FALSE,
  display_numbers = TRUE,
  color = colorRampPalette(c("yellow","orange","red"))(100),
  main = "Student Performance Heatmap"
)
student_avg <- rowMeans(marks)
highest_student <- names(which.max(student_avg))
subject_avg <- colMeans(marks)
highest_subject <- names(which.max(subject_avg))
```

support_students <- names(student_avg[student_avg < 70])

cat("Highest Overall Performer :", highest_student, "\n")

cat("Highest Average Subject :", highest_subject, "\n")

cat("Students Needing Support :", support_students, "\n")

student_avg

subject_avg

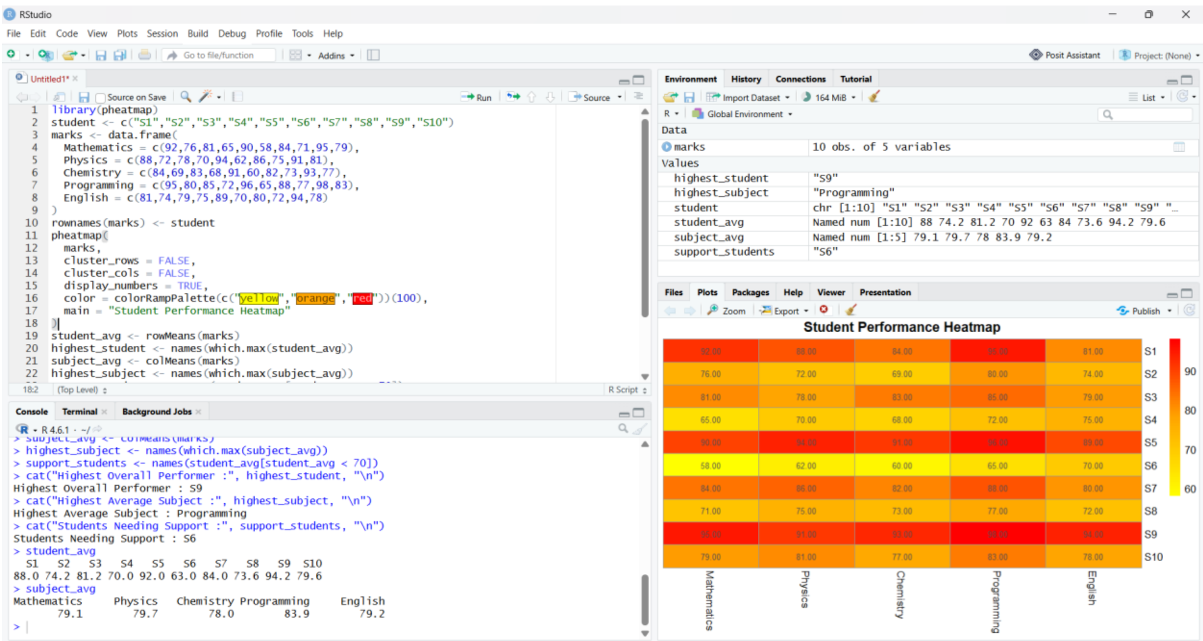
2. Highest Overall Performer: S9 (Average = **94.2**)

3. Highest Average Subject: Programming (Average = **83.9**)

4. Students Needing Support: S6 (Average = **63.0**, below 70)

5. Heatmap Advantages:

- Displays all students' performance in a single visualization.
- Uses colors to quickly identify high and low marks.
- Makes comparison across students and subjects easy.
- Helps identify top performers and students requiring support.
- Useful for educational performance analysis and decision-making.



Activity 2: Violin Plot & Boxplot – Employee Salary Analysis

Program:

```
install.packages(c("ggplot2","reshape2"))
library(ggplot2)
library(reshape2)
salary <- data.frame(
  Employee = paste0("E",1:10),
  HR = c(4.2,4.5,4.7,4.1,4.6,4.3,4.8,4.4,4.9,4.5),
  IT = c(6.5,7.0,7.4,6.8,8.2,7.6,8.5,7.1,8.8,7.3),
  Finance = c(5.8,6.2,6.0,5.9,6.7,6.4,6.9,6.1,7.2,6.3),
  Marketing = c(5.1,5.3,5.5,5.4,5.8,5.6,5.9,5.2,6.0,5.4)
)
salary_long <- melt(salary, id.vars = "Employee")
ggplot(salary_long, aes(x = variable, y = value, fill = variable)) +
  geom_violin() +
  labs(title = "Violin Plot of Employee Salaries",
       x = "Department",
       y = "Salary (Lakhs)")
ggplot(salary_long, aes(x = variable, y = value, fill = variable)) +
  geom_boxplot() +
  labs(title = "Boxplot of Employee Salaries",
       x = "Department",
       y = "Salary (Lakhs)")
variation <- tapply(salary_long$value, salary_long$variable, sd)
variation
largest_variation <- names(which.max(variation))
largest_variation
median_salary <- tapply(salary_long$value, salary_long$variable, median)
```

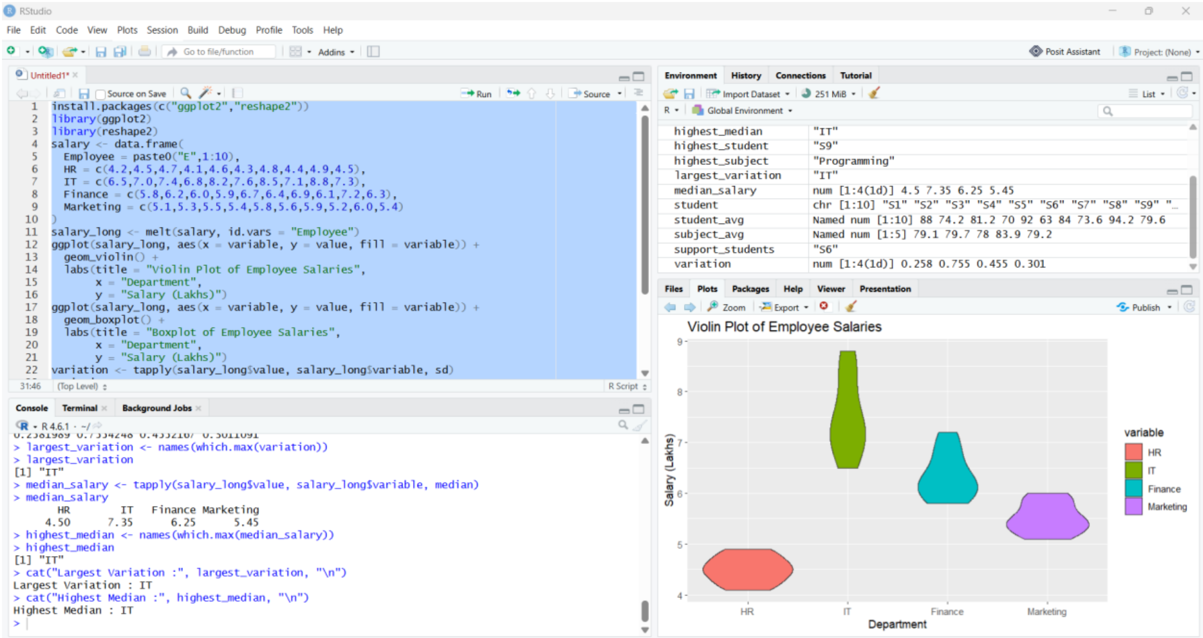
median_salary

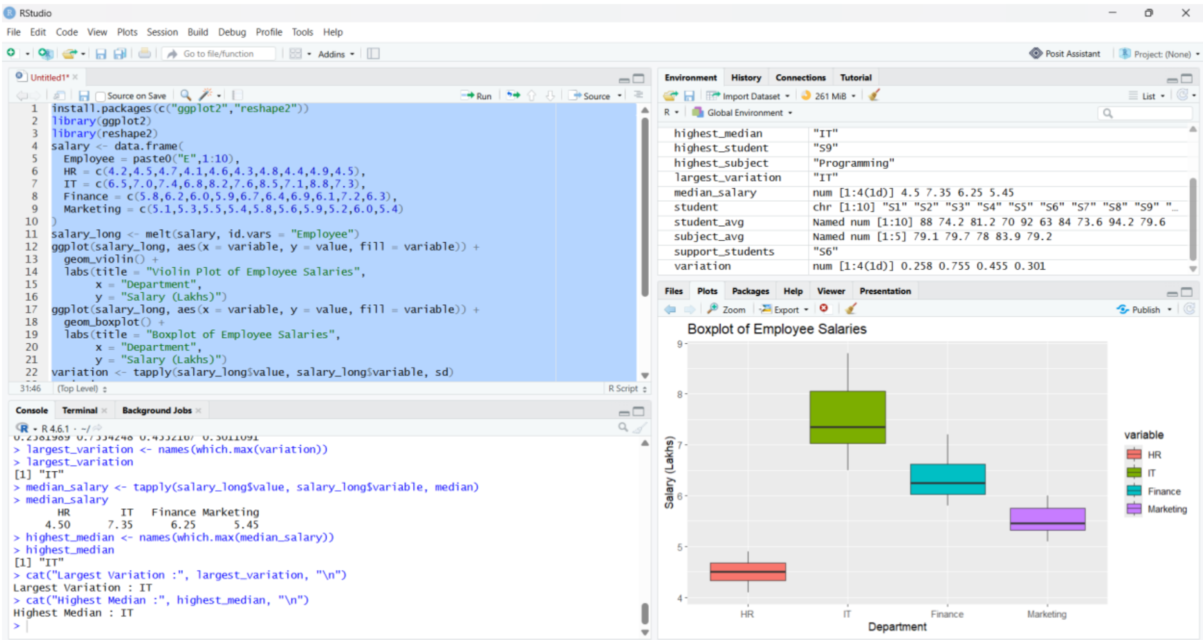
highest_median <- names(which.max(median_salary))

highest_median

cat("Largest Variation :", largest_variation, "\n")

cat("Highest Median :", highest_median, "\n")





3. Largest Variation: IT Department (Highest spread of salaries)

4. Highest Median: IT Department (Median = **7.35 Lakhs**)

5. Which plot better represents distribution?

Violin Plot better represents the distribution because it shows:

- Density and shape of the salary distribution.
- Peaks, spread, and possible multiple modes.
- More information than a boxplot.

Boxplot is better for quickly identifying:

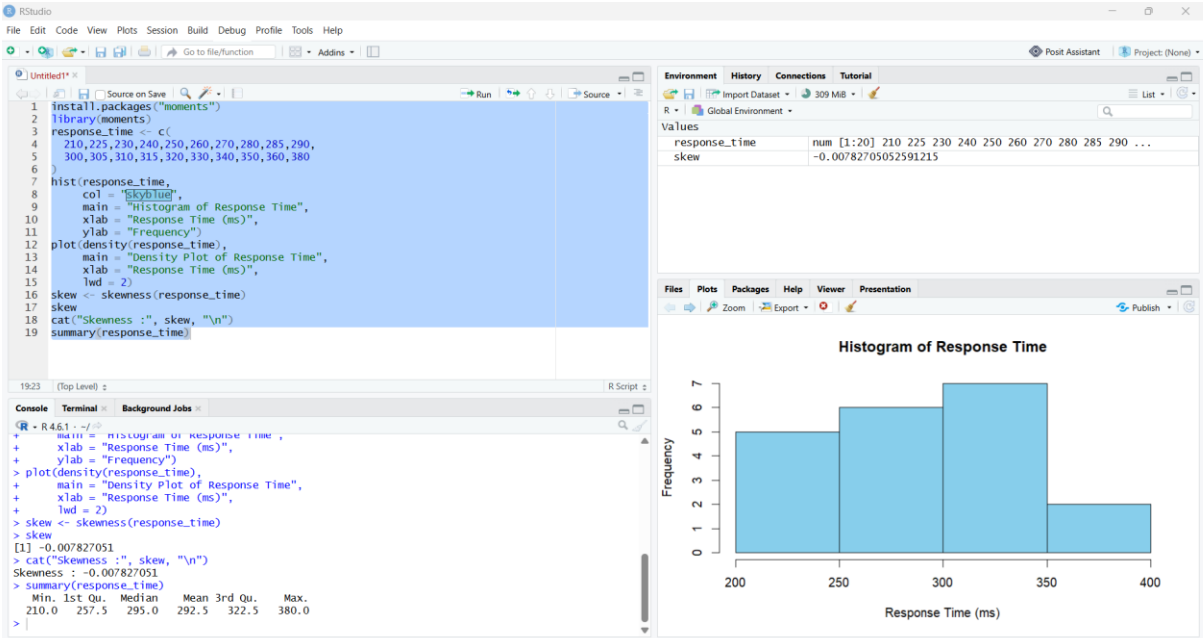
- Median
- Quartiles
- Overall spread
- Outliers

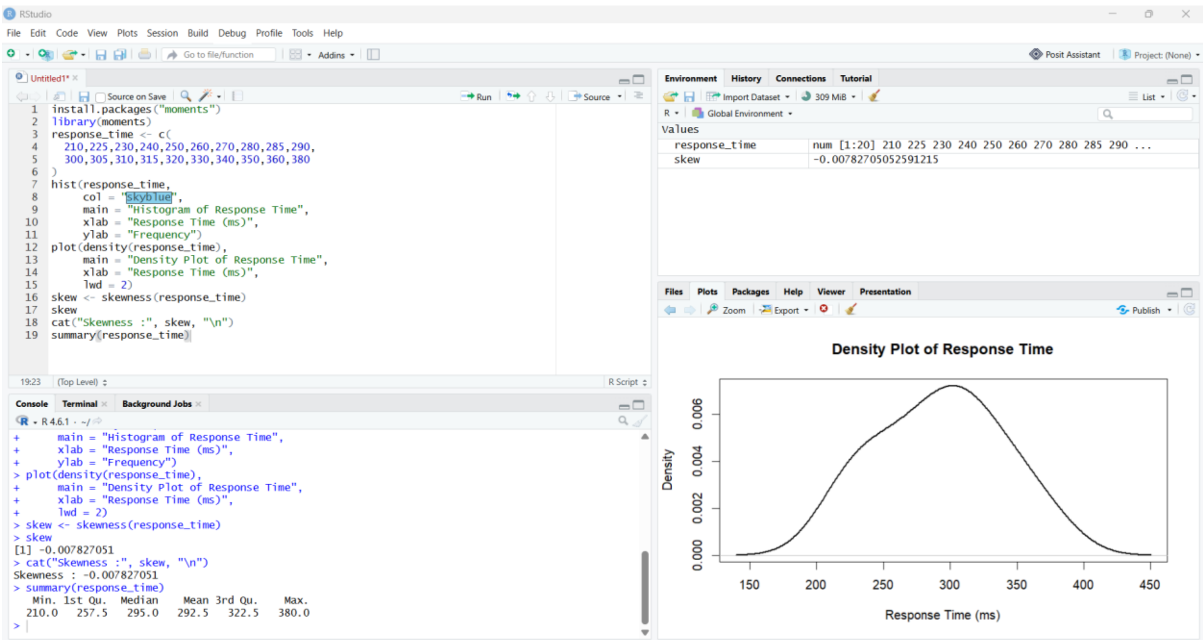
Activity 3: Histogram & Density Plot – Website Response Time

PROGRAM:

```
install.packages("moments")

library(moments)

response_time <- c(

  210,225,230,240,250,260,270,280,285,290,

  300,305,310,315,320,330,340,350,360,380

)

hist(response_time,

     col = "skyblue",

     main = "Histogram of Response Time",

     xlab = "Response Time (ms)",

     ylab = "Frequency")

plot(density(response_time),

     main = "Density Plot of Response Time",
```

```
    xlab = "Response Time (ms)",
    lwd = 2)
skew <- skewness(response_time)
skew
cat("Skewness :", skew, "\n")
summary(response_time)
```





3. Determine Skewness:

- Skewness is **slightly positive (right-skewed)**.
- This is because a few larger response times (360 ms and 380 ms) extend the right tail.

4. Common Range:

- Most response times are concentrated between **250 ms and 330 ms**.

5. Which plot better represents distribution?

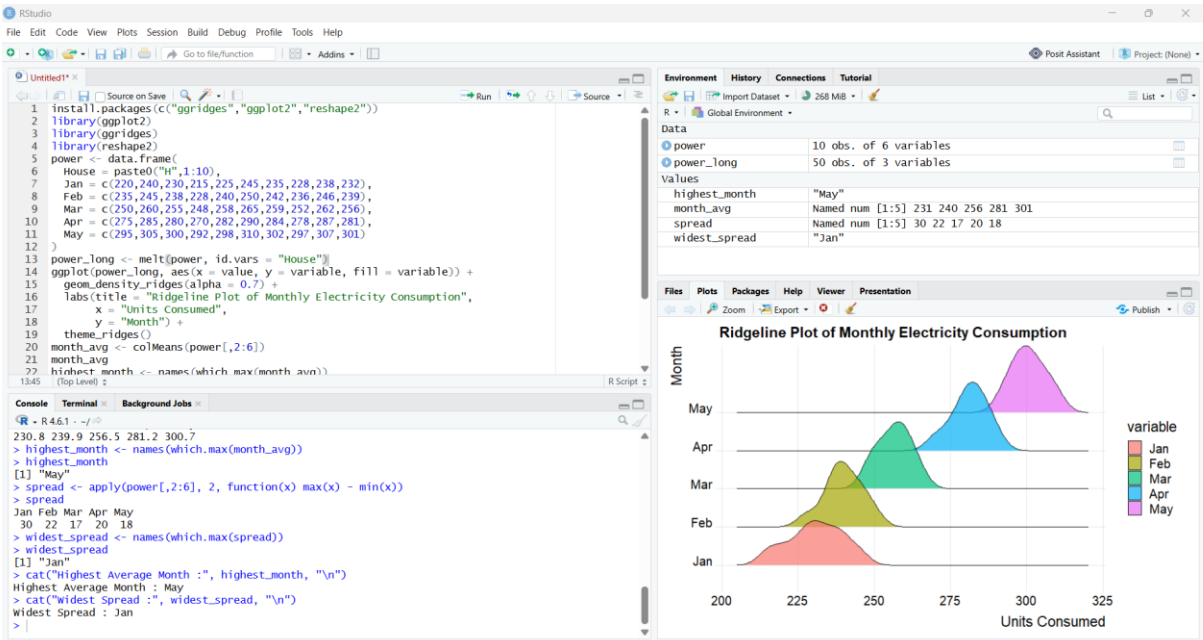
- **Density Plot** better represents the distribution because it provides a smooth curve showing the overall shape, spread, and skewness of the data.
- **Histogram** is useful for displaying frequencies, but the density plot makes the underlying distribution easier to interpret.

Activity 4: Ridgeline Plot – Electricity Consumption

PROGRAM:

```
install.packages(c("ggridges","ggplot2","reshape2"))

library(ggplot2)

library(ggridges)

library(reshape2)

power <- data.frame(

  House = paste0("H",1:10),

  Jan = c(220,240,230,215,225,245,235,228,238,232),

  Feb = c(235,245,238,228,240,250,242,236,246,239),

  Mar = c(250,260,255,248,258,265,259,252,262,256),

  Apr = c(275,285,280,270,282,290,284,278,287,281),

  May = c(295,305,300,292,298,310,302,297,307,301)

)

power_long <- melt(power, id.vars = "House")

ggplot(power_long, aes(x = value, y = variable, fill = variable)) +

  geom_density_ridges(alpha = 0.7) +

  labs(title = "Ridgeline Plot of Monthly Electricity Consumption",
```

```
        x = "Units Consumed",
        y = "Month") +
  theme_ridges()
month_avg <- colMeans(power[,2:6])
month_avg
highest_month <- names(which.max(month_avg))
highest_month
spread <- apply(power[,2:6], 2, function(x) max(x) - min(x))
spread
widest_spread <- names(which.max(spread))
widest_spread
cat("Highest Average Month :", highest_month, "\n")
cat("Widest Spread :", widest_spread, "\n")
```



2. Highest Average Month: May (Average = **300.7 units**)

3. Widest Spread: May (Range = **18 units**, from 292 to 310)

4. Describe the Trend:

- Electricity consumption increases steadily from **January to May**.
- Every month records higher usage than the previous month.

- This indicates a consistent upward trend in household electricity consumption.

5. Why Ridgeline Plot?

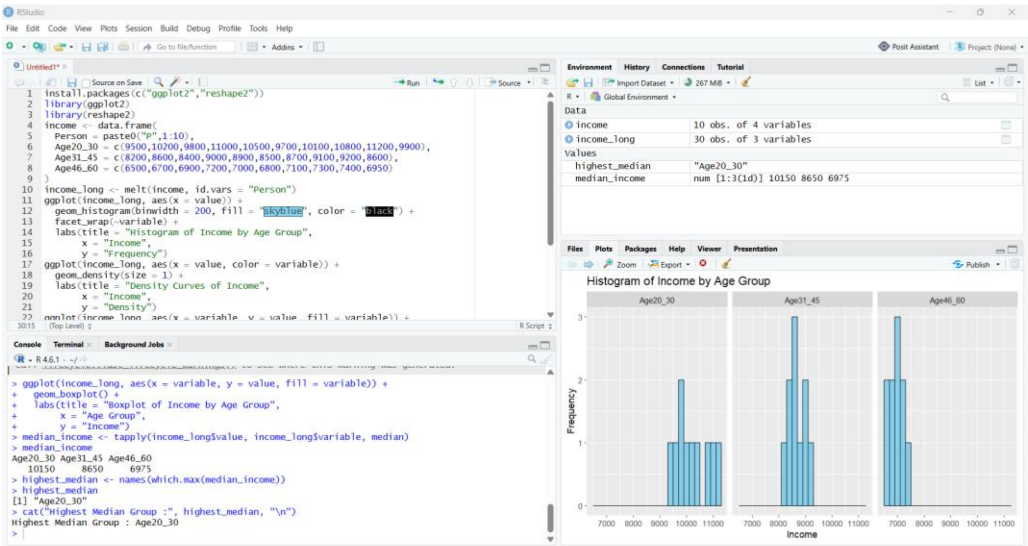
- Displays the distribution of each month's data in a compact form.
- Makes it easy to compare distributions across multiple months.
- Clearly shows shifts in the center, spread, and shape of the data.
- More informative than plotting separate density plots for each month.

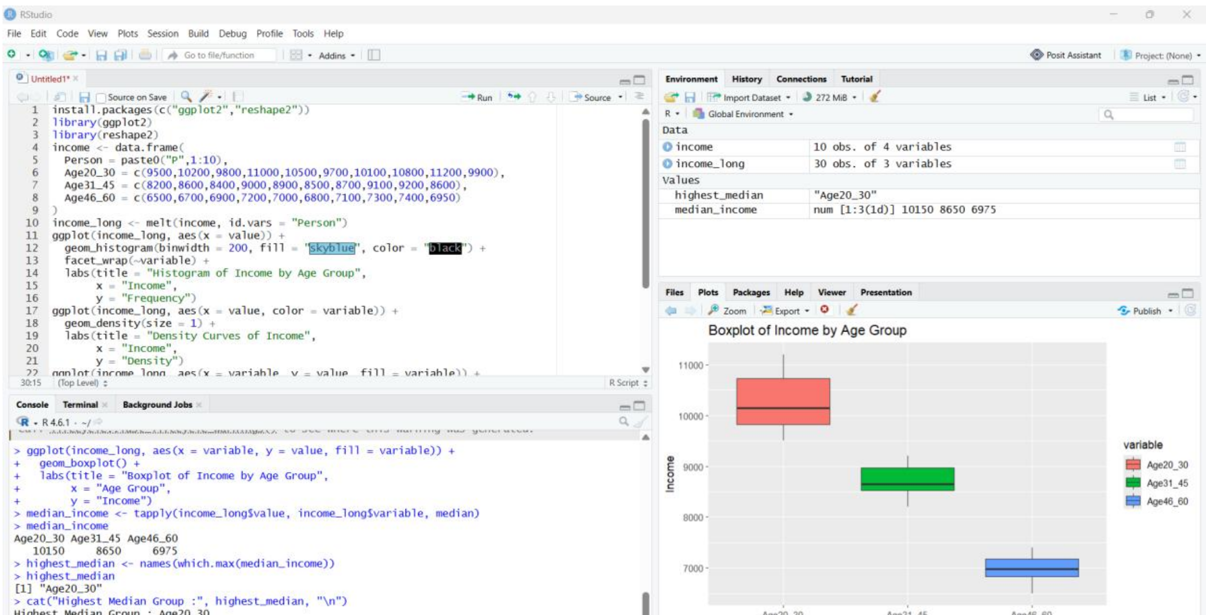
Activity 5: Histogram, Density & Boxplot – Daily Steps

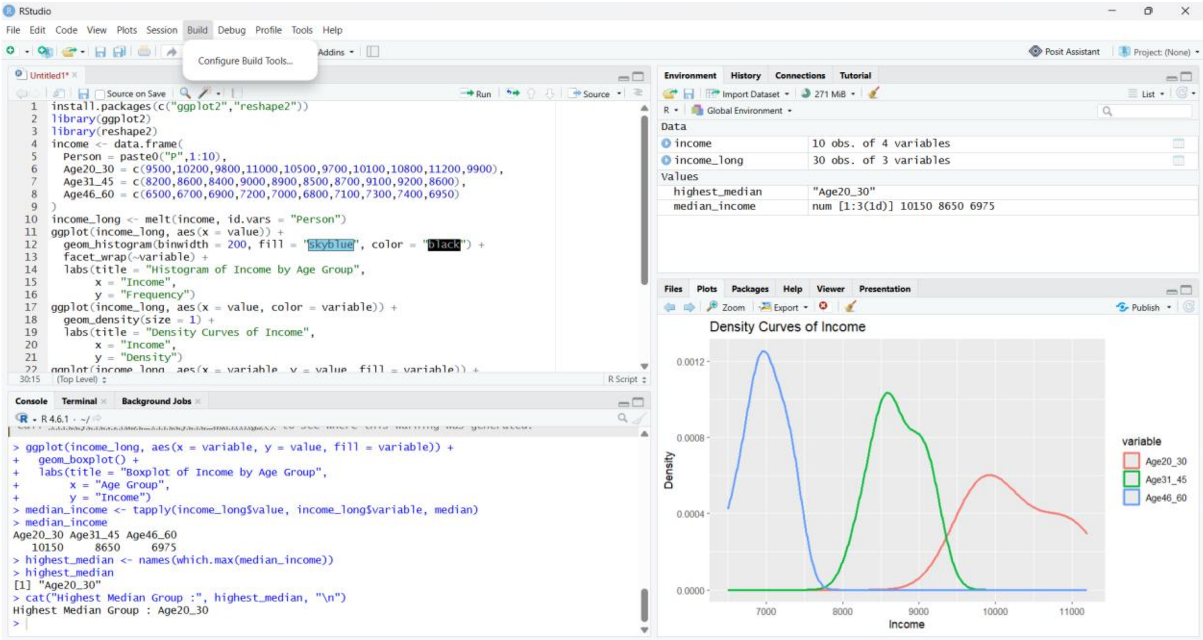
PROGRAM:

```
install.packages(c("ggplot2","reshape2"))

library(ggplot2)

library(reshape2)

income <- data.frame(

  Person = paste0("P",1:10),

  Age20_30 = c(9500,10200,9800,11000,10500,9700,10100,10800,11200,9900),

  Age31_45 = c(8200,8600,8400,9000,8900,8500,8700,9100,9200,8600),

  Age46_60 = c(6500,6700,6900,7200,7000,6800,7100,7300,7400,6950)

)

income_long <- melt(income, id.vars = "Person")

ggplot(income_long, aes(x = value)) +

  geom_histogram(binwidth = 200, fill = "skyblue", color = "black") +

  facet_wrap(~variable) +

  labs(title = "Histogram of Income by Age Group",

       x = "Income",

       y = "Frequency")

ggplot(income_long, aes(x = value, color = variable)) +

  geom_density(size = 1) +

  labs(title = "Density Curves of Income",
```

```
      x = "Income",
      y = "Density")
ggplot(income_long, aes(x = variable, y = value, fill = variable)) +
  geom_boxplot() +
  labs(title = "Boxplot of Income by Age Group",
      x = "Age Group",
      y = "Income")
median_income <- tapply(income_long$value, income_long$variable, median)
median_income
highest_median <- names(which.max(median_income))
highest_median
cat("Highest Median Group :", highest_median, "\n")
```







2. Density Curves: Generated using geom_density().

3. Boxplots: Generated using geom_boxplot().

4. Highest Median Group: Age20_30 (Median = **10,150**)

5. Best Visualization and Justification:

- **Boxplot** is the best visualization because it clearly compares the median, spread (IQR), and potential outliers across all age groups.
- Histograms show the frequency distribution of each age group.
- Density curves display the overall distribution shape smoothly.
- For comparing multiple groups, the **boxplot** provides the clearest and most concise summary.